Parameter priors and posteriors

The joint weekly dynamic stochastic correlation model

What this document shows. The prior specifies plausible parameter values before the estimation likelihood is applied. The posterior updates that distribution using the observed returns. This report gives the actual prior definitions, derives the conditional posterior updates, and compares the priors with saved parameter draws from this repository.

The current evidence is preliminary. The latest saved estimation contains **951 retained draws from four chains** (231, 238, 244, 238). The run stopped at its 12-hour time limit; its convergence record says `not_checked` because the diagnostic time budget was exhausted. The plots are therefore visualizations of saved draws intended to approximate the posterior, not a validated posterior estimate. No new MCMC run was launched to produce this document.

1. Which quantities are parameters?

For 14 weekly percentage log returns, with no VAR lags ($p = 0$), the observation model is

$$\begin{aligned} y_t \mid B_t, h_t, r_t &\sim \mathcal{N}(B_t, H_t), \\ H_t &= D_t P_t D_t, \qquad D_t = \operatorname{diag}\{\exp(h_{1t}/2), \ldots, \exp(h_{14,t}/2)\}, \\ r_t &= \operatorname{vecl}(\log P_t) \in \mathbb{R}^{91}. \end{aligned}$$

The logarithm here is a *matrix logarithm*; r_t is not a vector of ordinary correlations. The Archakov–Hansen inverse map produces the positive-definite correlation matrix P_t with unit diagonal [1].

Quantity	Meaning	Distinct entries
V	Covariance of weekly changes in expected returns B_t	105
$\sigma^2_{h,j}$	Variance of changes in log variance, one per series	14
$\sigma^2_{r,k}$	Variance of changes in a matrix-log coordinate	91
Fixed parameters	$\theta = (V, \sigma^2_h, \sigma^2_r)$	210
Weekly latent states	B_t, h_t, and r_t	119 per week

Interpretation. Larger evolution variances allow less smooth latent paths. The volatility parameter $\sigma^2_{h,j}$ is not the return variance $\exp(h_{jt})$. Likewise, V concerns changes in the mean, not the contemporaneous return covariance H_t.

2. What data are used?

The first 104 returns, 2003-08-15 to 2005-08-05, calibrate initial-state priors and are excluded from the likelihood. The selected estimation likelihood covers **1100 weeks**, 2005-08-12 to 2026-09-04. Evolution-prior scales use rolling windows through that estimation cutoff, including likelihood observations. This is empirical Bayes: the hyperparameters are fitted to data and subsequently held fixed. Their calibration uncertainty is not propagated. Historical state estimates are full-sample smoothers.

3. The priors actually used

Let $N_0 = 104$ be the complete-row count in the reserved calibration block, S_{ML} its maximum-likelihood covariance, and $S_0 = .95S_{\mathrm{ML}} + .05\,\mathrm{diag}(S_{\mathrm{ML}})$. The initial centers are

$$\bar{B} = \bar{y}_0, \qquad \bar{V}_B = S_0/N_0, \qquad \bar{h} = \log \mathrm{diag}(S_0), \qquad \bar{r} = \mathrm{vecl}\{\log \mathrm{Corr}(S_0)\}.$$

Mean states and their innovation covariance.

$$\begin{aligned} B_1 &\sim \mathcal{N}(\bar{B}, 4\bar{V}_B), \qquad B_1 \perp V, \\ B_t \mid B_{t-1}, V &\sim \mathcal{N}(B_{t-1}, V), \quad t = 2, \ldots, T, \\ V &\sim \mathrm{IW}(\Psi_0, \nu_0), \quad \Psi_0 = N_0\bar{V}_B(0.01)^2 = 10^{-4}S_0, \quad \nu_0 = 104. \end{aligned}$$

The convention is $p(V) \propto |V|^{-(\nu_0+m+1)/2} \exp\{-\frac{1}{2}\,\mathrm{tr}(\Psi_0 V^{-1})\}$, with $m = 14$. Thus $\mathrm{E}[V] = \Psi_0/89$, while the joint mode is $\Psi_0/119$ [2]. The diagonal marginal is $V_{jj} \sim \mathrm{IG}(45.5, \Psi_{0,jj}/2)$; the off-diagonal entries are jointly constrained by positive definiteness and may be negative.

Volatility states and their innovation variances. For each $j = 1, \ldots, 14$:

$$\begin{aligned} h_{j1} \mid \sigma^2_{h,j} &\sim \mathcal{N}(\bar{h}_j, 11\sigma^2_{h,j}), \\ h_{jt} \mid h_{j,t-1}, \sigma^2_{h,j} &\sim \mathcal{N}(h_{j,t-1}, \sigma^2_{h,j}), \quad t = 2, \ldots, T, \\ \sigma^2_{h,j} &\sim \mathrm{IG}(a, b_h), \quad a = 10, \quad b_h = 9.846 \times 10^{-3}. \end{aligned}$$

Correlation-coordinate states and their innovation variances. For each $k = 1, \ldots, 91$:

$$\begin{aligned} r_{k1} \mid \sigma^2_{r,k} &\sim \mathcal{N}(\bar{r}_k, 11\sigma^2_{r,k}), \\ r_{kt} \mid r_{k,t-1}, \sigma^2_{r,k} &\sim \mathcal{N}(r_{k,t-1}, \sigma^2_{r,k}), \quad t = 2, \ldots, T, \\ \sigma^2_{r,k} &\sim \mathrm{IG}(a, b_r), \quad a = 10, \quad b_r = 2.143 \times 10^{-3}. \end{aligned}$$

Each block follows the same order as B_t: first-state prior, law of motion, innovation-variance prior. Unlike $B_1 \perp V$, the priors for h_{j1} and r_{k1} depend on their innovation variances. Innovations are independent across dates and coordinates, conditional on their variances. The innovation-variance priors are independent within and across the two families. Here the *shape–scale* density and its moments are

$$p(s) = \frac{b^a}{\Gamma(a)} s^{-(a+1)} e^{-b/s}, \quad s > 0; \qquad \mathrm{E}[s] = \frac{b}{a-1}, \quad \mathrm{mode}(s) = \frac{b}{a+1}.$$

The calibrated means are $\mathrm{E}[\sigma^2_h] = 1.094 \times 10^{-3}$ and $\mathrm{E}[\sigma^2_r] = 2.381 \times 10^{-4}$. Each prior has coefficient of variation $1/\sqrt{a-2} = 0.354$. A common prior does not force the coordinate-specific posterior values to be equal. The scale b is not the prior mean.

How we obtain the priors for h_0 and $f_0 \equiv r_0$. This report denotes the correlation-coordinate state by r_t; equivalently, $f_0 = r_0$. From the same reserved 104-return block, form S_0 as above and set

$$D_S = \mathrm{diag}\{\sqrt{S_{0,11}}, \ldots, \sqrt{S_{0,14,14}}\}, \qquad C_0 = D_S^{-1} S_0 D_S^{-1}.$$

Then the coordinate-specific initial-state priors are

$$\begin{aligned} h_{j0} \mid \sigma^2_{h,j} &\sim \mathcal{N}(\bar{h}_j, 10\sigma^2_{h,j}), & \bar{h}_j &= \log S_{0,jj}, \\ r_{k0} \mid \sigma^2_{r,k} &\sim \mathcal{N}(\bar{r}_k, 10\sigma^2_{r,k}), & \bar{r}_k &= [\mathrm{vecl}(\log C_0)]_k. \end{aligned}$$

To calculate $\log C_0$, diagonalize $C_0 = Q\Lambda Q'$ and use $Q\,\mathrm{diag}(\log\lambda_1, \ldots, \log\lambda_{14})Q'$; retain its strict lower triangle in column-major order. Calibration fixes the centers $\bar{h}$ and $\bar{r}$; the pre-sample states h_0 and r_0 remain random, conditionally independent given their innovation variances.

For either coordinate z with innovation variance s and prior center μ, integrating out z_0 gives $\mathrm{Var}(z_1 \mid s) = 10s + s = 11s$, explaining the first-state priors above. The path covariance is $\mathrm{Cov}(z_t, z_u \mid s) = s\{10 + \min(t, u)\}$. The marginal prior for z_t is Student-t with 20 degrees of freedom, location μ, and scale $\sqrt{b(10+t)/a}$; its standard deviation is $\sqrt{(10+t)b/(a-1)}$.

How the evolution scales were calibrated. For each rolling window, the code differences rolling log variances and matrix-log coordinates, takes each coordinate's sample variance, then the median across coordinates. The selected window is 104 weeks.

Window (weeks)	Log-variance innovation mean	Coordinate innovation mean
52	3.553×10^{-3}	8.987×10^{-4}
104	1.094×10^{-3}	2.381×10^{-4}
156	4.885×10^{-4}	1.066×10^{-4}

4. From the prior to the posterior

Let $x = (B_{1:T}, h_{1:T}, r_{1:T})$ and let $\hat{\lambda}$ denote the fixed empirical-Bayes hyperparameters. Bayes' rule gives

$$p(\theta, x \mid y, \hat{\lambda}) \;\propto\; \underbrace{\prod_{t=1}^{T} \mathcal{N}(y_t; B_t, D_t P_t D_t)}_{\text{observation likelihood}} \quad \underbrace{p(x \mid \theta, \hat{\lambda}) p(\theta \mid \hat{\lambda})}_{\text{state and parameter priors}} \quad .$$

With missing values the likelihood uses the observed subvector and covariance submatrix. The desired parameter posterior is $p(\theta \mid y, \hat{\lambda}) = \int p(\theta, x \mid y, \hat{\lambda})\, dx$. This integration has no simple closed-form answer. MCMC approximates it by sampling states and parameters repeatedly.

Conditional posterior for V. Set $\Delta B_t = B_t - B_{t-1}$. Conditional on the sampled mean path,

$$\boxed{V \mid B_{1:T}, y, \ldots \sim \mathrm{IW}\left(\Psi_0 + \sum_{t=2}^{T} \Delta B_t \Delta B_t',\ \nu_0 + T - 1\right).}$$

For the selected $T = 1100$, the degrees of freedom are 1203. There are $T - 1$ increments because the B_1 prior is independent of V. The marginal posterior of V averages these inverse-Wishart conditionals over uncertain mean paths; it is generally a mixture, not a single inverse-Wishart distribution.

Conditional posterior for either innovation variance s. The integrated random-walk density contributes T Gaussian terms: the initial state with variance $11s$ and $T - 1$ changes with variance s. Therefore

$$\boxed{s \mid z_{1:T}, y, \ldots \sim \mathrm{IG}\left(a + \frac{T}{2},\ b + \frac{1}{2}\left[\frac{(z_1 - \mu)^2}{11} + \sum_{t=2}^{T} (z_t - z_{t-1})^2\right]\right).}$$

Use $(z, \mu, b) = (h_j, \bar{h}_j, b_h)$ or $(r_k, \bar{r}_k, b_r)$. The shape is 560 for this sample. The marginal posterior again integrates over uncertain paths; fitting one inverse-gamma density to all saved draws is unnecessary.

How this matches the sampler. The current `draw_variances` function uses an independence Metropolis–Hastings step. It proposes from $\mathrm{IG}(a + (T-1)/2, b + \frac{1}{2}\sum_{t=2}^{T} \Delta z_t^2)$ and accepts with probability

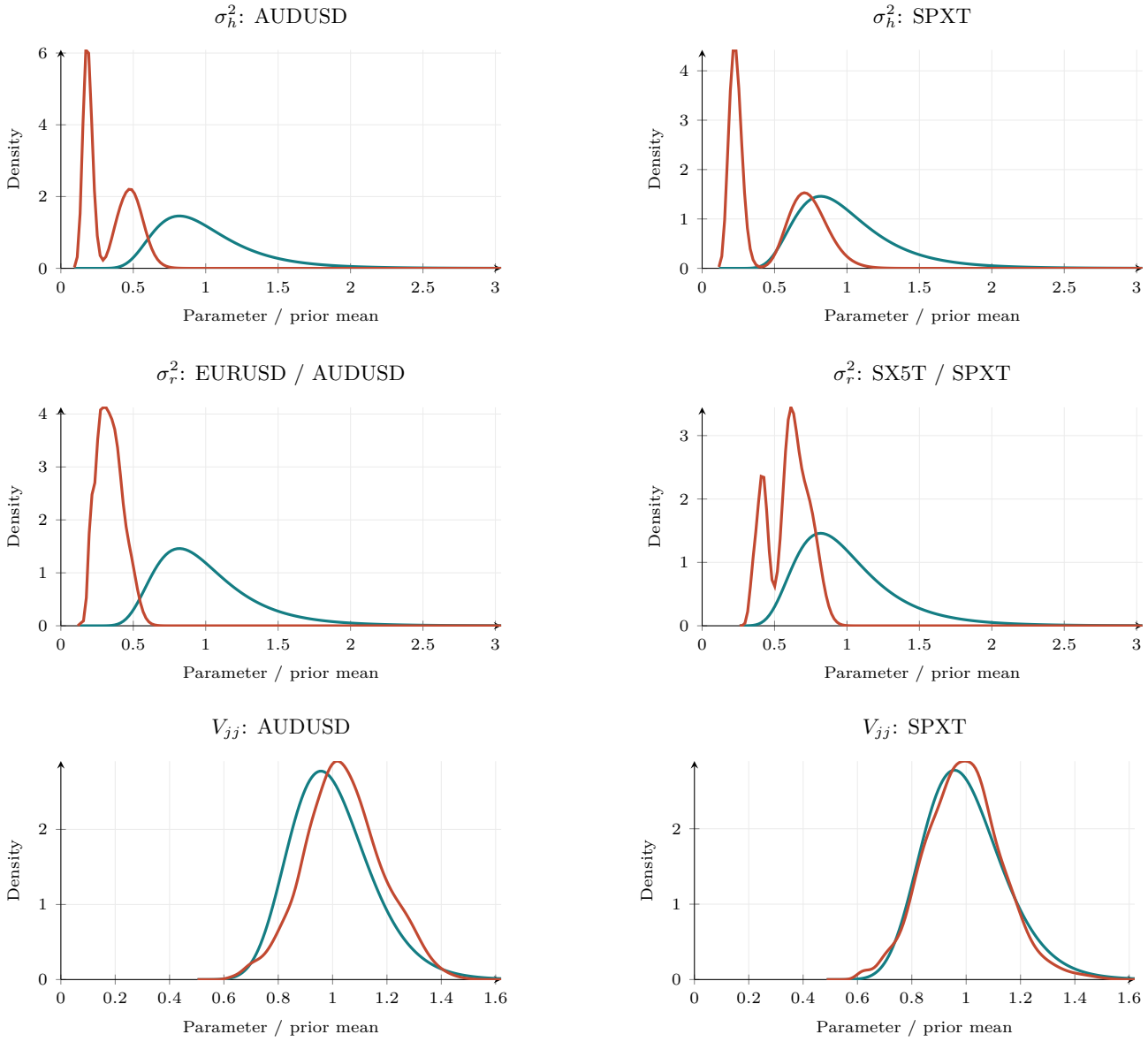
$$\min\left\{1, \left(\frac{s'}{s}\right)^{-1/2} \exp\left[-\frac{(z_1 - \mu)^2}{22}\left(\frac{1}{s'} - \frac{1}{s}\right)\right]\right\}.$$

That correction includes the initial-state density and targets the boxed conditional. The proposal distribution alone is not the full conditional posterior.

Conditional posteriors for the latent paths. Given h, r, V, $B_{1:T} \mid y, h, r, V$ is multivariate Gaussian and is sampled with a Kalman filter and backward simulation. The h_j and r_k path conditionals are non-Gaussian because they enter the covariance nonlinearly; their Gaussian random-walk priors are combined with the observation likelihood using elliptical slice updates [3]. Correlations and volatilities are obtained draw by draw as $P_t = \mathcal{C}(r_t)$ and $\exp(h_{jt}/2)$, before summarizing. In general, transforming a posterior mean is not the same as taking the posterior mean of a transformed quantity.

5. Prior versus saved-draw distributions

Teal: exact prior density. **Red: preliminary saved-draw density.** Each horizontal axis divides the parameter by its own prior mean; 1 is the prior mean. All 951 retained draws are pooled, so longer chains receive slightly more weight. The red line is a kernel density estimate on the log scale, transformed back to this positive scale with its Jacobian. These are marginal distributions, not state paths.



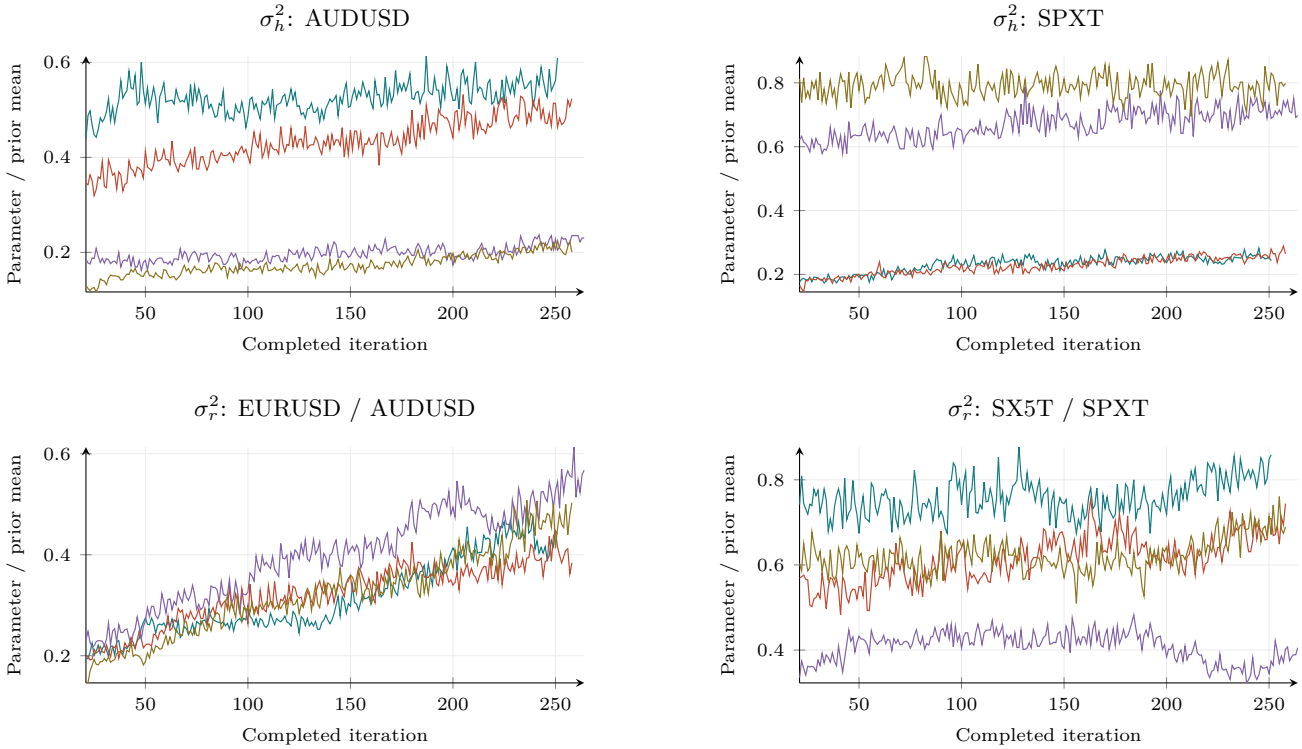
How to read a shift. A red curve to the left describes smaller evolution variances in the saved draws, hence smoother states. A narrower red curve describes less dispersion in these draws. Neither feature proves correct mixing or convergence. The two selected pairs illustrate familiar FX and equity labels; the appendix includes every fixed parameter, so the examples are not the full evidence.

6. What the current comparison can establish

The saved-draw means for all 14 log-variance innovation variances are 0.21–0.67 times their common prior mean. Across the 91 correlation-coordinate innovation variances, the corresponding ratios are 0.25–0.88. These describe the current chains; they are not established posterior findings.

Example	Prior mean	Saved-draw mean	Ratio
Volatility: AUDUSD	1.094×10^{-3}	3.608×10^{-4}	0.33
Volatility: SPXT	1.094×10^{-3}	5.285×10^{-4}	0.48
Coordinate: EURUSD / AUDUSD	2.381×10^{-4}	8.085×10^{-5}	0.34
Coordinate: SX5T / SPXT	2.381×10^{-4}	1.426×10^{-4}	0.60

Chain identity matters. The traces below preserve each chain and show only retained iterations after the configured 20-iteration burn-in. The colors denote chains 1, 2, 3, and 4. Similar pooled densities can conceal different chain locations.



Should the prior and posterior match? There is no requirement that they match, or that every posterior must be narrower. Similarity can mean weak information about a parameter or agreement with the prior. A shift may reflect learning, prior conflict, or sampling problems; its interpretation requires diagnostics and sensitivity checks.

Before treating draw percentiles as credible intervals: the repository's target is four independent chains, rank-normalized split/folded $\widehat{R} < 1.01$, bulk and tail ESS at least 400, and mean MCSE/posterior-SD at most 0.05 for reported quantities [4]. The saved diagnostic record contains no usable values for these metrics. The 20-iteration burn-in setting is a retention rule, not evidence that warm-up was sufficient. Passing diagnostics would still leave prior sensitivity and model adequacy to assess.

7. Initial-state priors and reproducibility

The following centers come from the reserved first 104 weekly returns. The B_1 entries are percentage points of weekly expected return; its prior SD is $2\sqrt{\bar{V}_{B,jj}}$. The log-variance centers use percentage-return units.

Series	$\bar{B}_j$	Prior SD of B_{j1}	$\bar{h}_j$
AUDUSD	0.1567	0.2993	0.8452
BCOMCOT	0.8553	0.8258	2.8753
BCOMGCTR	0.1873	0.3776	1.3105
EURUSD	0.0854	0.2618	0.5778
GBPUSD	0.1003	0.2405	0.4082
I02981JP	0.0061	0.0639	-2.2428
LEATTREU	0.1109	0.0846	-1.6824
LSG1TRGU	0.0880	0.1122	-1.1168
LUATTRUU	0.0656	0.1124	-1.1128
NKYTR	0.2423	0.4541	1.6793
SPXT	0.2518	0.2849	0.7470
SX5T	0.3214	0.3427	1.1163
TUKXG	0.3055	0.2423	0.4226
USDJPY	-0.0595	0.2421	0.4216

The B_1 prior is multivariate, with covariance $4\bar{V}_B$, not 14 independent normals. For h_1 and r_1, the marginal priors follow the Student-t expression on page 2. For $P_{ij,t}$ there is an induced joint prior obtained by mapping the entire r_t vector; assigning independent priors to ordinary pairwise correlations would change this model.

Fixed-parameter smoothing is a different output. When `estimate_parameters=false`, the workflow uses saved parameter means or saved parameter draws while updating latent states. This is not a new estimation of $p(\theta \mid y)$. Holding means fixed excludes parameter uncertainty; reusing saved draws includes their variation but does not reweight them into an updated parameter posterior when new observations arrive.

Saved evidence. Parameter file, estimated at 2026-09-27T09:08:55.845Z:

`dsc_parameters_20260927T090855845Z_0286d4ba-f106-4588-bd10-c52848277526.mat`

Run: `20260926-230854-651-chain1-w20-r500-thin1/estimation`.

Source SHA-256: `f11259e4c6347a54204cecb4f5ac3363c06ef2b05f787ab33ed9ab99e04c5bd7`. The companion numerical CSV contains prior moments and retained-draw means and quantiles for all 210 parameters. Its manifest records file hashes, chain counts, and validation of positive definiteness, draw counts, and agreement with the saved parameter means.

Implementation sources.

`toolbox/dsc_calibrate_priors.m`	Defines the priors
`toolbox/dsc_prepare_inference.m`	Splits calibration and likelihood dates
`toolbox/dsc_sample.m`	Implements the posterior updates
`toolbox/dsc_save_parameters.m`	Preserves draws and chain IDs
`scripts/build_parameter_explanation.py`	Generates this document

The generated LaTeX is self-contained and has no external figure dependencies.

Methodological references.

[1] Archakov and Hansen (2021). *A New Parametrization of Correlation Matrices*. Paper and reconstruction method.

[2] MathWorks. *iwishrnd*: inverse-Wishart parameterization and expectation. Official documentation.

[3] Murray, Adams and MacKay (2010). *Elliptical Slice Sampling*. PMLR 9, 541–548.

[4] Vehtari et al. (2021). *Rank-Normalization, Folding, and Localization*. Bayesian Analysis 16, 667–718.

Mean-evolution variances and volatility-evolution variances

All rows use the same saved run. Prior moments are analytical. The last three columns are empirical retained-draw summaries; convergence is unverified. Intervals are marginal 5th–95th percentiles, not simultaneous intervals. Ticker labels omit only the common “Index” suffix.

Parameter	Prior mean	Prior SD	Draw median	Draw 5%	Draw 95%
V(AUDUSD,AUDUSD)	2.616×10^{-6}	3.967×10^{-7}	2.711×10^{-6}	2.126×10^{-6}	3.352×10^{-6}
V(BCOMCOT,BCOMCOT)	1.992×10^{-5}	3.021×10^{-6}	1.982×10^{-5}	1.561×10^{-5}	2.433×10^{-5}
V(BCOMGCTR,BCOMGCTR)	4.166×10^{-6}	6.317×10^{-7}	4.271×10^{-6}	3.421×10^{-6}	5.322×10^{-6}
V(EURUSD,EURUSD)	2.002×10^{-6}	3.036×10^{-7}	2.070×10^{-6}	1.676×10^{-6}	2.582×10^{-6}
V(GBPUSD,GBPUSD)	1.690×10^{-6}	2.562×10^{-7}	1.678×10^{-6}	1.365×10^{-6}	2.087×10^{-6}
V(I02981JP,I02981JP)	1.193×10^{-7}	1.809×10^{-8}	1.195×10^{-7}	9.369×10^{-8}	1.602×10^{-7}
V(LEATTREU,LEATTREU)	2.089×10^{-7}	3.167×10^{-8}	2.103×10^{-7}	1.682×10^{-7}	2.785×10^{-7}
V(LSG1TRGU,LSG1TRGU)	3.678×10^{-7}	5.577×10^{-8}	3.673×10^{-7}	2.900×10^{-7}	4.827×10^{-7}
V(LUATTRUU,LUATTRUU)	3.692×10^{-7}	5.598×10^{-8}	3.851×10^{-7}	2.956×10^{-7}	4.892×10^{-7}
V(NKYTR,NKYTR)	6.024×10^{-6}	9.134×10^{-7}	5.886×10^{-6}	4.744×10^{-6}	7.297×10^{-6}
V(SPXT,SPXT)	2.372×10^{-6}	3.596×10^{-7}	2.349×10^{-6}	1.830×10^{-6}	2.868×10^{-6}
V(SX5T,SX5T)	3.431×10^{-6}	5.202×10^{-7}	3.344×10^{-6}	2.665×10^{-6}	4.315×10^{-6}
V(TUKXG,TUKXG)	1.715×10^{-6}	2.600×10^{-7}	1.665×10^{-6}	1.322×10^{-6}	2.065×10^{-6}
V(USDJPY,USDJPY)	1.713×10^{-6}	2.597×10^{-7}	1.762×10^{-6}	1.375×10^{-6}	2.323×10^{-6}
h(AUDUSD)	1.094×10^{-3}	3.868×10^{-4}	2.563×10^{-4}	1.718×10^{-4}	5.996×10^{-4}
h(BCOMCOT)	1.094×10^{-3}	3.868×10^{-4}	4.261×10^{-4}	1.978×10^{-4}	9.935×10^{-4}
h(BCOMGCTR)	1.094×10^{-3}	3.868×10^{-4}	4.070×10^{-4}	2.426×10^{-4}	1.034×10^{-3}
h(EURUSD)	1.094×10^{-3}	3.868×10^{-4}	6.085×10^{-4}	2.072×10^{-4}	1.092×10^{-3}
h(GBPUSD)	1.094×10^{-3}	3.868×10^{-4}	1.815×10^{-4}	9.270×10^{-5}	4.819×10^{-4}
h(I02981JP)	1.094×10^{-3}	3.868×10^{-4}	5.673×10^{-4}	2.052×10^{-4}	1.073×10^{-3}
h(LEATTREU)	1.094×10^{-3}	3.868×10^{-4}	6.006×10^{-4}	2.771×10^{-4}	9.441×10^{-4}
h(LSG1TRGU)	1.094×10^{-3}	3.868×10^{-4}	4.814×10^{-4}	2.229×10^{-4}	1.303×10^{-3}
h(LUATTRUU)	1.094×10^{-3}	3.868×10^{-4}	3.126×10^{-4}	1.640×10^{-4}	5.854×10^{-4}
h(NKYTR)	1.094×10^{-3}	3.868×10^{-4}	2.895×10^{-4}	1.318×10^{-4}	6.310×10^{-4}
h(SPXT)	1.094×10^{-3}	3.868×10^{-4}	6.486×10^{-4}	2.086×10^{-4}	8.986×10^{-4}
h(SX5T)	1.094×10^{-3}	3.868×10^{-4}	4.128×10^{-4}	2.993×10^{-4}	7.788×10^{-4}
h(TUKXG)	1.094×10^{-3}	3.868×10^{-4}	4.924×10^{-4}	2.825×10^{-4}	1.072×10^{-3}
h(USDJPY)	1.094×10^{-3}	3.868×10^{-4}	8.206×10^{-4}	3.353×10^{-4}	1.249×10^{-3}

V(i,j) describes covariance of changes in expected returns, in squared percentage-point units. h(j) denotes $\sigma^2_{h,j}$; r(i,j) denotes $\sigma^2_{r,k}$ for the matrix-log coordinate associated with that pair. Neither r(i,j) nor V(i,j) is a return correlation.

Correlation-coordinate evolution variances: 1–31

All rows use the same saved run. Prior moments are analytical. The last three columns are empirical retained-draw summaries; convergence is unverified. Intervals are marginal 5th–95th percentiles, not simultaneous intervals. Ticker labels omit only the common “Index” suffix.

Parameter	Prior mean	Prior SD	Draw median	Draw 5%	Draw 95%
r(BCOMCOT,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	7.210×10^{-5}	3.858×10^{-5}	1.215×10^{-4}
r(BCOMGCTR,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	9.311×10^{-5}	4.071×10^{-5}	1.843×10^{-4}
r(EURUSD,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	7.941×10^{-5}	4.949×10^{-5}	1.177×10^{-4}
r(GBPUSD,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	1.058×10^{-4}	6.456×10^{-5}	1.650×10^{-4}
r(I02981JP,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	7.898×10^{-5}	4.445×10^{-5}	1.053×10^{-4}
r(LEATTREU,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	1.214×10^{-4}	4.954×10^{-5}	1.877×10^{-4}
r(LSG1TRGU,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	7.803×10^{-5}	4.005×10^{-5}	1.161×10^{-4}
r(LUATTRUU,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	8.297×10^{-5}	4.644×10^{-5}	1.767×10^{-4}
r(NKYTR,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	7.064×10^{-5}	3.167×10^{-5}	9.763×10^{-5}
r(SPXT,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	1.625×10^{-4}	4.433×10^{-5}	3.964×10^{-4}
r(SX5T,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	1.163×10^{-4}	4.852×10^{-5}	1.861×10^{-4}
r(TUKXG,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	1.164×10^{-4}	6.985×10^{-5}	1.315×10^{-4}
r(USDJPY,AUDUSD)	2.381×10^{-4}	8.418×10^{-5}	1.441×10^{-4}	1.061×10^{-4}	1.849×10^{-4}
r(BCOMGCTR,BCOMCOT)	2.381×10^{-4}	8.418×10^{-5}	7.257×10^{-5}	3.788×10^{-5}	1.187×10^{-4}
r(EURUSD,BCOMCOT)	2.381×10^{-4}	8.418×10^{-5}	9.664×10^{-5}	4.494×10^{-5}	1.796×10^{-4}
r(GBPUSD,BCOMCOT)	2.381×10^{-4}	8.418×10^{-5}	8.811×10^{-5}	3.328×10^{-5}	1.107×10^{-4}
r(I02981JP,BCOMCOT)	2.381×10^{-4}	8.418×10^{-5}	8.414×10^{-5}	4.692×10^{-5}	1.210×10^{-4}
r(LEATTREU,BCOMCOT)	2.381×10^{-4}	8.418×10^{-5}	8.460×10^{-5}	4.079×10^{-5}	1.126×10^{-4}
r(LSG1TRGU,BCOMCOT)	2.381×10^{-4}	8.418×10^{-5}	1.197×10^{-4}	7.899×10^{-5}	1.475×10^{-4}
r(LUATTRUU,BCOMCOT)	2.381×10^{-4}	8.418×10^{-5}	1.359×10^{-4}	5.184×10^{-5}	1.905×10^{-4}
r(NKYTR,BCOMCOT)	2.381×10^{-4}	8.418×10^{-5}	6.757×10^{-5}	3.803×10^{-5}	1.083×10^{-4}
r(SPXT,BCOMCOT)	2.381×10^{-4}	8.418×10^{-5}	1.729×10^{-4}	1.280×10^{-4}	3.744×10^{-4}
r(SX5T,BCOMCOT)	2.381×10^{-4}	8.418×10^{-5}	8.058×10^{-5}	2.877×10^{-5}	1.613×10^{-4}
r(TUKXG,BCOMCOT)	2.381×10^{-4}	8.418×10^{-5}	8.809×10^{-5}	3.488×10^{-5}	1.277×10^{-4}
r(USDJPY,BCOMCOT)	2.381×10^{-4}	8.418×10^{-5}	7.041×10^{-5}	3.953×10^{-5}	1.816×10^{-4}
r(EURUSD,BCOMGCTR)	2.381×10^{-4}	8.418×10^{-5}	1.557×10^{-4}	5.210×10^{-5}	2.285×10^{-4}
r(GBPUSD,BCOMGCTR)	2.381×10^{-4}	8.418×10^{-5}	5.844×10^{-5}	2.875×10^{-5}	1.153×10^{-4}
r(I02981JP,BCOMGCTR)	2.381×10^{-4}	8.418×10^{-5}	7.230×10^{-5}	3.703×10^{-5}	9.722×10^{-5}
r(LEATTREU,BCOMGCTR)	2.381×10^{-4}	8.418×10^{-5}	8.622×10^{-5}	4.665×10^{-5}	1.038×10^{-4}
r(LSG1TRGU,BCOMGCTR)	2.381×10^{-4}	8.418×10^{-5}	7.485×10^{-5}	3.127×10^{-5}	1.650×10^{-4}
r(LUATTRUU,BCOMGCTR)	2.381×10^{-4}	8.418×10^{-5}	7.079×10^{-5}	3.704×10^{-5}	1.023×10^{-4}

V(i,j) describes covariance of changes in expected returns, in squared percentage-point units. h(j) denotes $\sigma^2_{h,j}$; r(i,j) denotes $\sigma^2_{r,k}$ for the matrix-log coordinate associated with that pair. Neither r(i,j) nor V(i,j) is a return correlation.

Correlation-coordinate evolution variances: 32–62

All rows use the same saved run. Prior moments are analytical. The last three columns are empirical retained-draw summaries; convergence is unverified. Intervals are marginal 5th–95th percentiles, not simultaneous intervals. Ticker labels omit only the common “Index” suffix.

Parameter	Prior mean	Prior SD	Draw median	Draw 5%	Draw 95%
r(NKYTR,BCOMGCTR)	2.381×10^{-4}	8.418×10^{-5}	7.867×10^{-5}	3.743×10^{-5}	1.447×10^{-4}
r(SPXT,BCOMGCTR)	2.381×10^{-4}	8.418×10^{-5}	6.517×10^{-5}	3.409×10^{-5}	8.672×10^{-5}
r(SX5T,BCOMGCTR)	2.381×10^{-4}	8.418×10^{-5}	7.917×10^{-5}	4.143×10^{-5}	1.383×10^{-4}
r(TUKXG,BCOMGCTR)	2.381×10^{-4}	8.418×10^{-5}	1.962×10^{-4}	4.067×10^{-5}	2.692×10^{-4}
r(USDJPY,BCOMGCTR)	2.381×10^{-4}	8.418×10^{-5}	7.348×10^{-5}	3.549×10^{-5}	2.047×10^{-4}
r(GBPUSD,EURUSD)	2.381×10^{-4}	8.418×10^{-5}	7.277×10^{-5}	3.072×10^{-5}	1.306×10^{-4}
r(I02981JP,EURUSD)	2.381×10^{-4}	8.418×10^{-5}	9.417×10^{-5}	4.656×10^{-5}	1.249×10^{-4}
r(LEATTREU,EURUSD)	2.381×10^{-4}	8.418×10^{-5}	8.373×10^{-5}	3.204×10^{-5}	1.192×10^{-4}
r(LSG1TRGU,EURUSD)	2.381×10^{-4}	8.418×10^{-5}	6.958×10^{-5}	3.351×10^{-5}	1.129×10^{-4}
r(LUATTRUU,EURUSD)	2.381×10^{-4}	8.418×10^{-5}	8.715×10^{-5}	4.701×10^{-5}	2.306×10^{-4}
r(NKYTR,EURUSD)	2.381×10^{-4}	8.418×10^{-5}	8.678×10^{-5}	5.284×10^{-5}	1.496×10^{-4}
r(SPXT,EURUSD)	2.381×10^{-4}	8.418×10^{-5}	1.432×10^{-4}	4.140×10^{-5}	2.130×10^{-4}
r(SX5T,EURUSD)	2.381×10^{-4}	8.418×10^{-5}	5.995×10^{-5}	3.006×10^{-5}	9.041×10^{-5}
r(TUKXG,EURUSD)	2.381×10^{-4}	8.418×10^{-5}	6.982×10^{-5}	3.613×10^{-5}	9.803×10^{-5}
r(USDJPY,EURUSD)	2.381×10^{-4}	8.418×10^{-5}	7.901×10^{-5}	3.410×10^{-5}	2.577×10^{-4}
r(I02981JP,GBPUSD)	2.381×10^{-4}	8.418×10^{-5}	7.307×10^{-5}	4.919×10^{-5}	1.056×10^{-4}
r(LEATTREU,GBPUSD)	2.381×10^{-4}	8.418×10^{-5}	7.223×10^{-5}	3.448×10^{-5}	1.802×10^{-4}
r(LSG1TRGU,GBPUSD)	2.381×10^{-4}	8.418×10^{-5}	6.943×10^{-5}	3.592×10^{-5}	1.024×10^{-4}
r(LUATTRUU,GBPUSD)	2.381×10^{-4}	8.418×10^{-5}	7.593×10^{-5}	4.152×10^{-5}	2.728×10^{-4}
r(NKYTR,GBPUSD)	2.381×10^{-4}	8.418×10^{-5}	1.753×10^{-4}	4.736×10^{-5}	4.452×10^{-4}
r(SPXT,GBPUSD)	2.381×10^{-4}	8.418×10^{-5}	9.080×10^{-5}	5.496×10^{-5}	2.126×10^{-4}
r(SX5T,GBPUSD)	2.381×10^{-4}	8.418×10^{-5}	8.453×10^{-5}	4.596×10^{-5}	2.236×10^{-4}
r(TUKXG,GBPUSD)	2.381×10^{-4}	8.418×10^{-5}	7.014×10^{-5}	3.937×10^{-5}	1.206×10^{-4}
r(USDJPY,GBPUSD)	2.381×10^{-4}	8.418×10^{-5}	1.296×10^{-4}	9.953×10^{-5}	1.527×10^{-4}
r(LEATTREU,I02981JP)	2.381×10^{-4}	8.418×10^{-5}	8.404×10^{-5}	3.892×10^{-5}	1.265×10^{-4}
r(LSG1TRGU,I02981JP)	2.381×10^{-4}	8.418×10^{-5}	6.385×10^{-5}	3.139×10^{-5}	9.797×10^{-5}
r(LUATTRUU,I02981JP)	2.381×10^{-4}	8.418×10^{-5}	8.499×10^{-5}	3.178×10^{-5}	1.954×10^{-4}
r(NKYTR,I02981JP)	2.381×10^{-4}	8.418×10^{-5}	1.629×10^{-4}	7.326×10^{-5}	2.723×10^{-4}
r(SPXT,I02981JP)	2.381×10^{-4}	8.418×10^{-5}	1.910×10^{-4}	5.715×10^{-5}	3.557×10^{-4}
r(SX5T,I02981JP)	2.381×10^{-4}	8.418×10^{-5}	8.153×10^{-5}	4.280×10^{-5}	9.892×10^{-5}
r(TUKXG,I02981JP)	2.381×10^{-4}	8.418×10^{-5}	7.064×10^{-5}	3.492×10^{-5}	1.256×10^{-4}

V(i,j) describes covariance of changes in expected returns, in squared percentage-point units. h(j) denotes $\sigma^2_{h,j}$; r(i,j) denotes $\sigma^2_{r,k}$ for the matrix-log coordinate associated with that pair. Neither r(i,j) nor V(i,j) is a return correlation.

Correlation-coordinate evolution variances: 63–91

All rows use the same saved run. Prior moments are analytical. The last three columns are empirical retained-draw summaries; convergence is unverified. Intervals are marginal 5th–95th percentiles, not simultaneous intervals. Ticker labels omit only the common “Index” suffix.

Parameter	Prior mean	Prior SD	Draw median	Draw 5%	Draw 95%
r(USDJPY,I02981JP)	2.381×10^{-4}	8.418×10^{-5}	7.389×10^{-5}	4.249×10^{-5}	1.074×10^{-4}
r(LSG1TRGU,LEATTREU)	2.381×10^{-4}	8.418×10^{-5}	7.885×10^{-5}	2.685×10^{-5}	1.609×10^{-4}
r(LUATTRUU,LEATTREU)	2.381×10^{-4}	8.418×10^{-5}	1.003×10^{-4}	4.283×10^{-5}	1.895×10^{-4}
r(NKYTR,LEATTREU)	2.381×10^{-4}	8.418×10^{-5}	1.096×10^{-4}	8.115×10^{-5}	1.329×10^{-4}
r(SPXT,LEATTREU)	2.381×10^{-4}	8.418×10^{-5}	1.166×10^{-4}	3.763×10^{-5}	1.631×10^{-4}
r(SX5T,LEATTREU)	2.381×10^{-4}	8.418×10^{-5}	7.585×10^{-5}	4.013×10^{-5}	1.233×10^{-4}
r(TUKXG,LEATTREU)	2.381×10^{-4}	8.418×10^{-5}	1.538×10^{-4}	6.964×10^{-5}	3.654×10^{-4}
r(USDJPY,LEATTREU)	2.381×10^{-4}	8.418×10^{-5}	7.045×10^{-5}	3.373×10^{-5}	1.139×10^{-4}
r(LUATTRUU,LSG1TRGU)	2.381×10^{-4}	8.418×10^{-5}	1.042×10^{-4}	4.782×10^{-5}	2.029×10^{-4}
r(NKYTR,LSG1TRGU)	2.381×10^{-4}	8.418×10^{-5}	6.295×10^{-5}	3.825×10^{-5}	1.048×10^{-4}
r(SPXT,LSG1TRGU)	2.381×10^{-4}	8.418×10^{-5}	1.061×10^{-4}	2.742×10^{-5}	1.502×10^{-4}
r(SX5T,LSG1TRGU)	2.381×10^{-4}	8.418×10^{-5}	7.337×10^{-5}	4.337×10^{-5}	1.065×10^{-4}
r(TUKXG,LSG1TRGU)	2.381×10^{-4}	8.418×10^{-5}	6.854×10^{-5}	2.678×10^{-5}	1.089×10^{-4}
r(USDJPY,LSG1TRGU)	2.381×10^{-4}	8.418×10^{-5}	8.565×10^{-5}	4.076×10^{-5}	1.514×10^{-4}
r(NKYTR,LUATTRUU)	2.381×10^{-4}	8.418×10^{-5}	8.236×10^{-5}	5.160×10^{-5}	2.665×10^{-4}
r(SPXT,LUATTRUU)	2.381×10^{-4}	8.418×10^{-5}	6.787×10^{-5}	3.737×10^{-5}	1.054×10^{-4}
r(SX5T,LUATTRUU)	2.381×10^{-4}	8.418×10^{-5}	1.029×10^{-4}	5.068×10^{-5}	1.792×10^{-4}
r(TUKXG,LUATTRUU)	2.381×10^{-4}	8.418×10^{-5}	1.472×10^{-4}	6.291×10^{-5}	3.718×10^{-4}
r(USDJPY,LUATTRUU)	2.381×10^{-4}	8.418×10^{-5}	7.537×10^{-5}	3.983×10^{-5}	2.055×10^{-4}
r(SPXT,NKYTR)	2.381×10^{-4}	8.418×10^{-5}	7.147×10^{-5}	3.337×10^{-5}	2.723×10^{-4}
r(SX5T,NKYTR)	2.381×10^{-4}	8.418×10^{-5}	1.312×10^{-4}	4.065×10^{-5}	2.842×10^{-4}
r(TUKXG,NKYTR)	2.381×10^{-4}	8.418×10^{-5}	1.195×10^{-4}	6.680×10^{-5}	1.642×10^{-4}
r(USDJPY,NKYTR)	2.381×10^{-4}	8.418×10^{-5}	1.378×10^{-4}	8.049×10^{-5}	2.073×10^{-4}
r(SX5T,SPXT)	2.381×10^{-4}	8.418×10^{-5}	1.470×10^{-4}	8.895×10^{-5}	1.892×10^{-4}
r(TUKXG,SPXT)	2.381×10^{-4}	8.418×10^{-5}	1.087×10^{-4}	5.279×10^{-5}	1.471×10^{-4}
r(USDJPY,SPXT)	2.381×10^{-4}	8.418×10^{-5}	1.060×10^{-4}	3.749×10^{-5}	1.674×10^{-4}
r(TUKXG,SX5T)	2.381×10^{-4}	8.418×10^{-5}	1.013×10^{-4}	4.900×10^{-5}	2.826×10^{-4}
r(USDJPY,SX5T)	2.381×10^{-4}	8.418×10^{-5}	1.089×10^{-4}	4.376×10^{-5}	1.372×10^{-4}
r(USDJPY,TUKXG)	2.381×10^{-4}	8.418×10^{-5}	6.682×10^{-5}	3.591×10^{-5}	1.061×10^{-4}

V(i,j) describes covariance of changes in expected returns, in squared percentage-point units. h(j) denotes $\sigma^2_{h,j}$; r(i,j) denotes $\sigma^2_{r,k}$ for the matrix-log coordinate associated with that pair. Neither r(i,j) nor V(i,j) is a return correlation.

Mean-evolution covariances: 1–31

All rows use the same saved run. Prior moments are analytical. The last three columns are empirical retained-draw summaries; convergence is unverified. Intervals are marginal 5th–95th percentiles, not simultaneous intervals. Ticker labels omit only the common “Index” suffix.

Parameter	Prior mean	Prior SD	Draw median	Draw 5%	Draw 95%
V(BCOMCOT,AUDUSD)	1.380×10^{-6}	7.840×10^{-7}	1.452×10^{-6}	4.410×10^{-7}	3.188×10^{-6}
V(BCOMGCTR,AUDUSD)	2.114×10^{-6}	4.193×10^{-7}	2.217×10^{-6}	1.597×10^{-6}	2.891×10^{-6}
V(EURUSD,AUDUSD)	1.385×10^{-6}	2.861×10^{-7}	1.476×10^{-6}	1.047×10^{-6}	1.869×10^{-6}
V(GBPUSD,AUDUSD)	1.298×10^{-6}	2.642×10^{-7}	1.367×10^{-6}	9.588×10^{-7}	1.768×10^{-6}
V(I02981JP,AUDUSD)	-2.598×10^{-8}	5.963×10^{-8}	-1.408×10^{-8}	-1.026×10^{-7}	8.572×10^{-8}
V(LEATTREU,AUDUSD)	3.686×10^{-8}	7.892×10^{-8}	6.150×10^{-8}	-8.117×10^{-8}	1.974×10^{-7}
V(LSG1TRGU,AUDUSD)	9.384×10^{-9}	1.046×10^{-7}	4.492×10^{-8}	-1.213×10^{-7}	1.895×10^{-7}
V(LUATTRUU,AUDUSD)	1.192×10^{-7}	1.056×10^{-7}	1.713×10^{-7}	-1.604×10^{-9}	3.587×10^{-7}
V(NKYTR,AUDUSD)	7.726×10^{-7}	4.314×10^{-7}	7.269×10^{-7}	-1.020×10^{-7}	1.423×10^{-6}
V(SPXT,AUDUSD)	5.340×10^{-7}	2.717×10^{-7}	5.625×10^{-7}	1.189×10^{-7}	9.827×10^{-7}
V(SX5T,AUDUSD)	7.740×10^{-8}	3.195×10^{-7}	7.368×10^{-8}	-4.291×10^{-7}	5.312×10^{-7}
V(TUKXG,AUDUSD)	-4.667×10^{-8}	2.259×10^{-7}	-5.600×10^{-8}	-4.136×10^{-7}	3.084×10^{-7}
V(USDJPY,AUDUSD)	-1.093×10^{-6}	2.546×10^{-7}	-1.146×10^{-6}	-1.614×10^{-6}	-7.712×10^{-7}
V(BCOMGCTR,BCOMCOT)	1.118×10^{-6}	9.788×10^{-7}	1.149×10^{-6}	-1.850×10^{-7}	2.908×10^{-6}
V(EURUSD,BCOMCOT)	5.954×10^{-7}	6.764×10^{-7}	7.366×10^{-7}	-2.650×10^{-7}	1.739×10^{-6}
V(GBPUSD,BCOMCOT)	5.599×10^{-7}	6.216×10^{-7}	6.975×10^{-7}	-1.352×10^{-7}	1.749×10^{-6}
V(I02981JP,BCOMCOT)	-7.716×10^{-8}	1.646×10^{-7}	-1.333×10^{-7}	-3.971×10^{-7}	1.451×10^{-7}
V(LEATTREU,BCOMCOT)	1.100×10^{-7}	2.178×10^{-7}	1.176×10^{-7}	-2.844×10^{-7}	5.401×10^{-7}
V(LSG1TRGU,BCOMCOT)	2.361×10^{-7}	2.897×10^{-7}	2.527×10^{-7}	-2.326×10^{-7}	8.178×10^{-7}
V(LUATTRUU,BCOMCOT)	5.173×10^{-8}	2.892×10^{-7}	4.875×10^{-8}	-4.549×10^{-7}	5.951×10^{-7}
V(NKYTR,BCOMCOT)	-7.481×10^{-7}	1.171×10^{-6}	-4.387×10^{-7}	-2.477×10^{-6}	1.663×10^{-6}
V(SPXT,BCOMCOT)	-1.154×10^{-6}	7.433×10^{-7}	-8.794×10^{-7}	-1.918×10^{-6}	2.511×10^{-7}
V(SX5T,BCOMCOT)	-1.388×10^{-6}	8.940×10^{-7}	-1.103×10^{-6}	-2.427×10^{-6}	1.596×10^{-7}
V(TUKXG,BCOMCOT)	-6.862×10^{-7}	6.275×10^{-7}	-5.787×10^{-7}	-1.490×10^{-6}	4.298×10^{-7}
V(USDJPY,BCOMCOT)	-4.050×10^{-8}	6.228×10^{-7}	-5.303×10^{-8}	-9.656×10^{-7}	8.047×10^{-7}
V(EURUSD,BCOMGCTR)	1.644×10^{-6}	3.553×10^{-7}	1.723×10^{-6}	1.179×10^{-6}	2.309×10^{-6}
V(GBPUSD,BCOMGCTR)	1.527×10^{-6}	3.273×10^{-7}	1.572×10^{-6}	1.069×10^{-6}	2.011×10^{-6}
V(I02981JP,BCOMGCTR)	-6.664×10^{-8}	7.550×10^{-8}	-5.996×10^{-8}	-1.799×10^{-7}	6.820×10^{-8}
V(LEATTREU,BCOMGCTR)	9.527×10^{-8}	9.999×10^{-8}	1.125×10^{-7}	-4.958×10^{-8}	3.017×10^{-7}
V(LSG1TRGU,BCOMGCTR)	7.326×10^{-8}	1.322×10^{-7}	1.207×10^{-7}	-1.064×10^{-7}	3.016×10^{-7}
V(LUATTRUU,BCOMGCTR)	2.478×10^{-7}	1.349×10^{-7}	2.966×10^{-7}	7.357×10^{-8}	5.333×10^{-7}

V(i,j) describes covariance of changes in expected returns, in squared percentage-point units. h(j) denotes $\sigma^2_{h,j}$; r(i,j) denotes $\sigma^2_{r,k}$ for the matrix-log coordinate associated with that pair. Neither r(i,j) nor V(i,j) is a return correlation.

Mean-evolution covariances: 32–62

All rows use the same saved run. Prior moments are analytical. The last three columns are empirical retained-draw summaries; convergence is unverified. Intervals are marginal 5th–95th percentiles, not simultaneous intervals. Ticker labels omit only the common “Index” suffix.

Parameter	Prior mean	Prior SD	Draw median	Draw 5%	Draw 95%
V(NKYTR,BCOMGCTR)	5.849×10^{-7}	5.378×10^{-7}	5.856×10^{-7}	-2.856×10^{-7}	1.388×10^{-6}
V(SPXT,BCOMGCTR)	3.054×10^{-7}	3.367×10^{-7}	2.903×10^{-7}	-1.755×10^{-7}	7.787×10^{-7}
V(SX5T,BCOMGCTR)	-3.405×10^{-7}	4.047×10^{-7}	-3.769×10^{-7}	-9.189×10^{-7}	1.807×10^{-7}
V(TUKXG,BCOMGCTR)	-3.722×10^{-7}	2.878×10^{-7}	-4.023×10^{-7}	-8.069×10^{-7}	-1.701×10^{-9}
V(USDJPY,BCOMGCTR)	-1.429×10^{-6}	3.238×10^{-7}	-1.493×10^{-6}	-2.042×10^{-6}	-9.655×10^{-7}
V(GBPUSD,EURUSD)	1.304×10^{-6}	2.413×10^{-7}	1.363×10^{-6}	1.056×10^{-6}	1.747×10^{-6}
V(I02981JP,EURUSD)	5.497×10^{-8}	5.244×10^{-8}	6.618×10^{-8}	-7.884×10^{-9}	1.758×10^{-7}
V(LEATTREU,EURUSD)	7.135×10^{-9}	6.896×10^{-8}	1.944×10^{-8}	-9.279×10^{-8}	1.350×10^{-7}
V(LSG1TRGU,EURUSD)	6.993×10^{-9}	9.150×10^{-8}	2.862×10^{-8}	-1.033×10^{-7}	2.019×10^{-7}
V(LUATTRUU,EURUSD)	8.399×10^{-8}	9.212×10^{-8}	1.143×10^{-7}	-5.504×10^{-8}	2.655×10^{-7}
V(NKYTR,EURUSD)	-3.584×10^{-7}	3.723×10^{-7}	-4.343×10^{-7}	-1.016×10^{-6}	9.291×10^{-8}
V(SPXT,EURUSD)	1.988×10^{-7}	2.333×10^{-7}	2.189×10^{-7}	-1.197×10^{-7}	6.558×10^{-7}
V(SX5T,EURUSD)	-5.268×10^{-7}	2.851×10^{-7}	-5.283×10^{-7}	-9.280×10^{-7}	-1.326×10^{-7}
V(TUKXG,EURUSD)	-2.613×10^{-7}	1.995×10^{-7}	-2.784×10^{-7}	-5.580×10^{-7}	1.171×10^{-8}
V(USDJPY,EURUSD)	-7.076×10^{-7}	2.117×10^{-7}	-7.454×10^{-7}	-1.108×10^{-6}	-4.494×10^{-7}
V(I02981JP,GBPUSD)	1.265×10^{-8}	4.789×10^{-8}	1.726×10^{-8}	-4.229×10^{-8}	1.020×10^{-7}
V(LEATTREU,GBPUSD)	3.927×10^{-8}	6.349×10^{-8}	4.439×10^{-8}	-6.127×10^{-8}	1.585×10^{-7}
V(LSG1TRGU,GBPUSD)	-1.974×10^{-8}	8.408×10^{-8}	-8.385×10^{-9}	-1.435×10^{-7}	1.201×10^{-7}
V(LUATTRUU,GBPUSD)	1.284×10^{-7}	8.535×10^{-8}	1.375×10^{-7}	1.747×10^{-9}	3.035×10^{-7}
V(NKYTR,GBPUSD)	9.333×10^{-9}	3.402×10^{-7}	-1.714×10^{-8}	-6.289×10^{-7}	4.605×10^{-7}
V(SPXT,GBPUSD)	1.342×10^{-7}	2.139×10^{-7}	1.842×10^{-7}	-1.669×10^{-7}	5.318×10^{-7}
V(SX5T,GBPUSD)	-3.626×10^{-7}	2.597×10^{-7}	-3.254×10^{-7}	-7.200×10^{-7}	-1.476×10^{-8}
V(TUKXG,GBPUSD)	-3.893×10^{-7}	1.863×10^{-7}	-3.595×10^{-7}	-6.126×10^{-7}	-1.237×10^{-7}
V(USDJPY,GBPUSD)	-7.109×10^{-7}	1.969×10^{-7}	-7.207×10^{-7}	-1.079×10^{-6}	-4.417×10^{-7}
V(LEATTREU,I02981JP)	4.393×10^{-8}	1.748×10^{-8}	3.878×10^{-8}	1.000×10^{-8}	7.459×10^{-8}
V(LSG1TRGU,I02981JP)	5.043×10^{-8}	2.298×10^{-8}	4.516×10^{-8}	1.046×10^{-8}	9.211×10^{-8}
V(LUATTRUU,I02981JP)	5.584×10^{-8}	2.317×10^{-8}	5.293×10^{-8}	1.205×10^{-8}	1.054×10^{-7}
V(NKYTR,I02981JP)	-3.962×10^{-7}	9.996×10^{-8}	-3.907×10^{-7}	-5.859×10^{-7}	-2.361×10^{-7}
V(SPXT,I02981JP)	-1.216×10^{-7}	5.820×10^{-8}	-1.054×10^{-7}	-2.012×10^{-7}	-1.686×10^{-8}
V(SX5T,I02981JP)	-1.877×10^{-7}	7.114×10^{-8}	-1.716×10^{-7}	-2.825×10^{-7}	-7.382×10^{-8}
V(TUKXG,I02981JP)	-1.138×10^{-7}	4.975×10^{-8}	-1.079×10^{-7}	-1.862×10^{-7}	-4.446×10^{-8}

V(i,j) describes covariance of changes in expected returns, in squared percentage-point units. h(j) denotes $\sigma^2_{h,j}$; r(i,j) denotes $\sigma^2_{r,k}$ for the matrix-log coordinate associated with that pair. Neither r(i,j) nor V(i,j) is a return correlation.

Mean-evolution covariances: 63–91

All rows use the same saved run. Prior moments are analytical. The last three columns are empirical retained-draw summaries; convergence is unverified. Intervals are marginal 5th–95th percentiles, not simultaneous intervals. Ticker labels omit only the common “Index” suffix.

Parameter	Prior mean	Prior SD	Draw median	Draw 5%	Draw 95%
V(USDJPY,I02981JP)	-1.139×10^{-8}	4.820×10^{-8}	-2.148×10^{-8}	-1.203×10^{-7}	4.584×10^{-8}
V(LSG1TRGU,LEATTREU)	2.348×10^{-7}	3.891×10^{-8}	2.352×10^{-7}	1.834×10^{-7}	3.179×10^{-7}
V(LUATTRUU,LEATTREU)	1.980×10^{-7}	3.650×10^{-8}	2.081×10^{-7}	1.366×10^{-7}	2.707×10^{-7}
V(NKYTR,LEATTREU)	-2.640×10^{-7}	1.229×10^{-7}	-2.297×10^{-7}	-4.503×10^{-7}	-1.683×10^{-8}
V(SPXT,LEATTREU)	-2.202×10^{-7}	7.871×10^{-8}	-2.129×10^{-7}	-3.416×10^{-7}	-1.003×10^{-7}
V(SX5T,LEATTREU)	-2.941×10^{-7}	9.567×10^{-8}	-2.993×10^{-7}	-4.973×10^{-7}	-1.503×10^{-7}
V(TUKXG,LEATTREU)	-2.273×10^{-7}	6.835×10^{-8}	-2.312×10^{-7}	-3.609×10^{-7}	-1.199×10^{-7}
V(USDJPY,LEATTREU)	5.216×10^{-9}	6.377×10^{-8}	-2.694×10^{-8}	-1.366×10^{-7}	9.101×10^{-8}
V(LUATTRUU,LSG1TRGU)	2.423×10^{-7}	4.718×10^{-8}	2.523×10^{-7}	1.697×10^{-7}	3.356×10^{-7}
V(NKYTR,LSG1TRGU)	-3.389×10^{-7}	1.628×10^{-7}	-2.976×10^{-7}	-6.371×10^{-7}	-4.242×10^{-8}
V(SPXT,LSG1TRGU)	-2.562×10^{-7}	1.033×10^{-7}	-2.525×10^{-7}	-4.129×10^{-7}	-9.895×10^{-8}
V(SX5T,LSG1TRGU)	-3.408×10^{-7}	1.253×10^{-7}	-3.459×10^{-7}	-5.634×10^{-7}	-1.679×10^{-7}
V(TUKXG,LSG1TRGU)	-2.560×10^{-7}	8.905×10^{-8}	-2.572×10^{-7}	-4.197×10^{-7}	-1.355×10^{-7}
V(USDJPY,LSG1TRGU)	3.554×10^{-8}	8.470×10^{-8}	-6.752×10^{-9}	-1.397×10^{-7}	1.364×10^{-7}
V(NKYTR,LUATTRUU)	-3.579×10^{-7}	1.636×10^{-7}	-3.366×10^{-7}	-6.068×10^{-7}	-4.539×10^{-8}
V(SPXT,LUATTRUU)	-2.701×10^{-7}	1.039×10^{-7}	-2.627×10^{-7}	-4.498×10^{-7}	-1.254×10^{-7}
V(SX5T,LUATTRUU)	-4.163×10^{-7}	1.281×10^{-7}	-4.265×10^{-7}	-6.603×10^{-7}	-2.472×10^{-7}
V(TUKXG,LUATTRUU)	-3.561×10^{-7}	9.311×10^{-8}	-3.598×10^{-7}	-5.273×10^{-7}	-2.094×10^{-7}
V(USDJPY,LUATTRUU)	-1.061×10^{-7}	8.555×10^{-8}	-1.455×10^{-7}	-3.142×10^{-7}	-6.931×10^{-9}
V(SPXT,NKYTR)	1.859×10^{-6}	4.501×10^{-7}	1.762×10^{-6}	1.108×10^{-6}	2.512×10^{-6}
V(SX5T,NKYTR)	2.754×10^{-6}	5.684×10^{-7}	2.666×10^{-6}	1.880×10^{-6}	3.599×10^{-6}
V(TUKXG,NKYTR)	1.683×10^{-6}	3.877×10^{-7}	1.622×10^{-6}	1.138×10^{-6}	2.263×10^{-6}
V(USDJPY,NKYTR)	-2.963×10^{-7}	3.439×10^{-7}	-2.788×10^{-7}	-7.600×10^{-7}	2.961×10^{-7}
V(SX5T,SPXT)	2.054×10^{-6}	3.762×10^{-7}	2.008×10^{-6}	1.533×10^{-6}	2.653×10^{-6}
V(TUKXG,SPXT)	1.229×10^{-6}	2.525×10^{-7}	1.219×10^{-6}	8.724×10^{-7}	1.677×10^{-6}
V(USDJPY,SPXT)	-1.652×10^{-7}	2.156×10^{-7}	-1.411×10^{-7}	-4.779×10^{-7}	2.266×10^{-7}
V(TUKXG,SX5T)	1.945×10^{-6}	3.329×10^{-7}	1.916×10^{-6}	1.494×10^{-6}	2.517×10^{-6}
V(USDJPY,SX5T)	2.339×10^{-7}	2.597×10^{-7}	2.523×10^{-7}	-1.390×10^{-7}	6.139×10^{-7}
V(USDJPY,TUKXG)	3.118×10^{-7}	1.858×10^{-7}	3.169×10^{-7}	5.682×10^{-8}	6.328×10^{-7}

V(i,j) describes covariance of changes in expected returns, in squared percentage-point units. h(j) denotes $\sigma^2_{h,j}$; r(i,j) denotes $\sigma^2_{r,k}$ for the matrix-log coordinate associated with that pair. Neither r(i,j) nor V(i,j) is a return correlation.